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MINIMON: A soft monitor for the 7768

MON 1 board

MINIMON (mini-monitor) is a 13 command monitor that enables the 7768 to be attached to an asynchronous terminal via the 7768 MON 1 board. MINIMON can be loaded from cassette or paper tape into the memory on the MON 1 board (FC00-FFFF) using the bootstrap loader function.

The monitor supports the following commands:

1. X,Z Relative offset calculation
2. M Examine and modify memory
3. A Alter memory
4. G Go to user program and run it
5. S Set break point
6. C Continue running a program after a break point
7. R Print 6800 registers
8. H Gives headings to print out by command R
9. B Block move
10. D Dumps a block of data to the terminal
11. P Punch unformatted binary cassette or paper tape
12. L Load from tape produced by the punch command

Issue 1 : £4.50 including cassette VAT 8%



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MINIMON: A soft monitor for the 7768
MON 1 board.

INTRODUCTION

MINIMON is a simple 1K monitor that allows the interface of the 7768 with an asynchronous data terminal and provides a method of entering, debugging and saving programs.

It occupies the 1K space on the MON 1 board (FC00-FFFF) and uses the locations F0DD-F0FE (on the CPU board) for temporary storage. The stack used by MINIMON (may also be used by user programs) extends from F0D1 Downwards.

The ASCII send/receive terminal (assumed to operate in a full duplex mode) must be connected to ACIA (a) , or ACIA (b) if locations FC18, FD34 and FF26 are changed from 00 to 02, and locations FC11, FD05, FD2D, FF2C and FF96 are changed from 01 to 03.

USE

MINIMON can be loaded using the MON 1 Bootstrap Loader function. It is then initiated by hitting the reset button. MINIMON will respond with an asterisk and will then wait for one of the following commands:

1. X,Z Relative offset calculation
2. M Examine and modify memory
3. A Alter memory
4. G Go-run user program
5. S Set break point
6. C Continue running a program after a break point
7. R Register print-6800 registers printed
8. H Header-gives titles to print out by function R
9. B Block move

- 10. D Dumps a block of data to the terminal
- 11. P Punch-unformatted binary dump
- 12. L Load from tape produced by the punch comand

COMMANDS

Z (Relative offset calculation)

This command enables the calculation of branch offsets to be done automatically. MINIMON is supplied with the address of the first byte of the brach instruction and the address of destination and the offset is then calculated according to the formula:

$$\text{OFFSET} = \text{DESTINATION} - (\text{PC} + 2)$$

Format (user input underlined)

*Z P aa D bb R=cc

where aa is the least significant two HEX digits of the branch instruction address, bb is the LSB of the destination address. cc is the offset.

Attempts to branch more than 127 bytes forward or 128 bytes back are indicated by RANGE! being printed.

X (Relative offset calculation)

This command is similar to Z except that the addresses are the full 16 bit, 4 HEX digits to be typed. eg:

*XP 00F0 D 0110 R=1E

A(Alter)

If A is typed while the monitor is waiting for a comand ('*'printed) MINIMON will reply with a space which is a prompt for an address. The user should type the address (in HEX) of the first location to be altered. The monitor will then print this address on the next line, The user can then type in the new contents of that location. The command can then be terminated by pressing full stop, or the next memory location

can be altered by typing two HEX digits.

FORMAT

*A 0000

0000 aabbccdd...

Where aa, bb, cc, dd etc. are the data entered into locations 0000, 0001, 0002, 0003 etc. respectively.

When the end of a line is reached MINIMON will print the current address at the beginning of the next line and the command is continued.

M (Modify)

The modify command can be used for both examining and altering memory. The function is entered by typing an M, MINIMON will respond with a space and then wait for the HEX address to be typed in. The monitor will print the contents of this location and then wait for further inputs which may be any of the following:

- 1 A space which will cause the monitor to print the contents of the next location.
- 2 A slash '/' which will cause the monitor to print the contents of the previous location and its address on the next line.
- 3 A carriage return (CR) which will cause MINIMON to print the contents of the next location and its address on the next line.
- 4 Two HEX digits to be stored in the location whose contents was last printed.

FORMAT

*M 0000 aa_bb/

0000 aa (CR)

0001 bbcc dd

Where aa, bb are the contents of 0000, 0001 respectively. cc was entered into location 0001 in place of bb.

When the end of a line is reached MINIMON will continue the modify function at the beginning of the next line.

The modify function, as with all commands, can be left at any time by typing a full stop.

G (Go)

The G function enables the user program to be entered at the specified address and run from that point. After the G command has been entered the user should type the HEX address of the start of the program. This can then be checked and if it is incorrect type a full stop to exit the function (with no harm done). If it is correct any character (apart from a full stop) will cause the system to run the user program until it encounters a SWI (3F) or the system reset button is pressed.

FORMAT

*G aaaa Where aaaa is the address of the start of the user program. Any character was then typed to run the program.

S (Set break point)

Using this command a program can be stopped at any predetermined point. The user selects this point and types in its address. MINIMON will then save the instruction at that point and replace it with a SWI (3F). Two break points may be set at any one time.

FORMAT

*S1 aaaa Where aaaa is the address of break point 1
OR

*S2 bbbb Where bbbb is the address of break point 2

C (Continue)

After a program has been interrupted (using the S command) registers, memory, etc. may be examined and altered. Then to set the program running again from the point it left off type either C1 or C2 according to which break point was encountered.

FORMAT

*C1

OR

*C2

R (Register print)

If a break point (SWI) is encountered while a user program is running, the 6800 registers will be pushed on to the stack, and control will be passed to MINIMON. Using the R command the registers can then be printed from the stack.

FORMAT

*R CC B A X PC SP

Where CC=Condition Codes, B and A are the accumulators, X=Index reg., PC=Program Counter, SP=Stack Pointer.

These registers can then be altered on the stack using the Modify command, the HEX address is SP+1.

FORMAT

*R CF 0B 34 0004 0017 F0D0

*M F0D1 CF 0B00 34 00 04.

Where acc. B was changed from 0B to 00.

The values of these registers (Altered or not) will then be reloaded into the 6800 MPU when the next G or C command is given.

H (Header)

This command gives titles to the print out by the register print instruction.

FORMAT

*H CC B A X PC SP

*R CF 0B 34 0004 0017 F0D0

B (Block Move)

Using the Block move function a block of data or program of any size can be moved to any location. For example.

The block of data with starting address 0100 and finishing address 0110 could be moved one location up, so that its new starting address would be 0101. This could be done as follows:

*B S 0100 F 0110 TO 0101

where S=Start address of block, F=Finnish address of block, and the last entry is the new start address of the block. (All entries are in HEX)

NOTE: Block move can also be used to move a block (say 0100 to 0110) one or more locations down (say to the new starting address 00FF) as follows:

*B S 0100 F 0110 TO 00FF

In each case, although the original block will be overwritten, the relocated block will be faithfully reproduced.

D (Dump)

MINIMON will dump the contents of memory between two specified addresses to the terminal. There are 8 (typically but can be adjusted-see listing) Bytes per line with the address of the first Byte on each line printed first. The dump can be stopped at any time by typing any character at the terminal keyboard.

FORMAT

*D S aaaa F bbbb

aaaa cc cc cc cc ...

Where aaaa is the address of the first Byte, bbbb the add. of the last Byte of the area of data to be printed. cc was the data found.

PUNCH

Program or data can be dumped (unformatted binary) onto cassette or paper tape using this command. Note MINIMON will punch two ASCII S's at the beginning of each data dump as a start of file indicator.

FORMAT

*P S aaaa F bbbb

Where aaaa, bbbb are the boundary addresses.



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HEXADECIMAL CODING FORM

PROGRAM <i>MINIMON</i>		VERSION		AUTHOR <i>ACH</i>		DATE <i>DEC. '78</i>	PAGE <i>1</i> OF <i>4</i>
ADDRESS	MACHINE CODE	LABEL	OPERATOR & OPERAND		COMMENTS		
		CTRLA	EQU	\$F401	ACIA(a) CTRL/STATUS		
		CTRLB	EQU	\$F403	ACIA(b) " "		
		DATAA	EQU	\$F400	ACIA(a) DATA REG.		
		DATAB	EQU	\$F402	ACIA(b) " "		
		STACK	EQU	\$F000	TOP OF MINIMON'S STACK		
		JNMI	EQU	\$F000	SPACE FOR JUMP TO NMI SUB.		
		JIRQ	EQU	\$F0E0	" " " " IRQ "		
		SAVE	EQU	\$F0E3	TEMP. STORAGE FOR NEWLINE		
		P	EQU	\$F0E5	" " " Z,X		
		X	EQU	\$F0E6	" " " SUB		
		Y	EQU	\$F0E8	" " " SUB		
		NEW	EQU	\$F0EA	" " " BLOCK MOVE		
		M	EQU	\$F0EC	" " " ZOUT		
		TEMPX	EQU	\$F0E0	" " " PRX		
		START	EQU	\$F0EF	" " " GETADDRESS		
		STOP	EQU	\$F0F1	" " " GETADDRESS		
		Z	EQU	\$F0F3	" " " . RDX		
		R	EQU	\$F0F5	" " " PR		
		Q	EQU	\$F0F6	" " " ZIN, DUMP		
		ABYTE	EQU	\$F0F7	" " " SET BR. PT.		
		BYTE	EQU	\$F0FB	" " " SET BR. PT.		
		PSTACK	EQU	\$F0F0	" " " SWI		
<i>F.C.00</i>	<i>B0.F0.2B</i>	RD	JSR	RDB	INPUT ONE CHARACTER		
<i>03</i>	<i>27.FB</i>		BEQ	RD	IGNOR PAPER TAPE FOLLOWER		
<i>05</i>	<i>80.08</i>		BSR	PR	ECHO CHARACTER		
<i>07</i>	<i>81.2E</i>		CMP	A #'.	WAS CHARACTER A FULLSTOP		
<i>09</i>	<i>26.03</i>		BNE	END			
<i>0B</i>	<i>7E.FF.8F</i>		JMP	START	YES - GOTO START OF MINIMON		
<i>0E</i>	<i>39</i>	END	RTS		NO - RETURN		
<i>0F</i>	<i>FF.6F.40</i>	PR	LOA	B CTRLA	GET ACIA(a) CTRL BYTE		
<i>12</i>	<i>C5.02</i>		BIT	B #02	IS ACIA(a) BUSSY?		
<i>14</i>	<i>27.F9</i>		BEQ	PR	YES - TRY AGAIN		
<i>16</i>	<i>B2.F4.00</i>		STA	A DATAA	NO - SEND DATA		
<i>19</i>	<i>7C.F0.F5</i>		INC	R	CHARACTERS PRINTED + ONE		
<i>1C</i>	<i>39</i>		RTS		RETURN TO CALLER		



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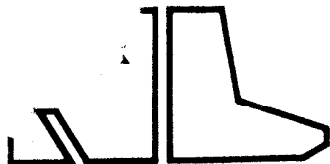
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PROGRAM MINIMON		VERSION		AUTHOR ACH	DATE DEC '78	PAGE 2 OF 4
ADDRESS	MACHINE CODE	LABEL	OPERATOR & OPERAND		COMMENTS	
F.C.1.D	8.1.2.F	VHEX	CMP A #2F		A < 30;	
1.F	2.F0E		BLE NO		NOT HEX	
21	8.1.3.9		CMP A #39		A > 39;	
23	2.2.0.2		BHI N		NOT A NUMERAL	
25	0.C	YES	CLC		ITS OK	
26	3.9		RTS		SO RETURN	
27	8.1.4.0	N	CMP A #40		A < 41	
29	2.F04		BLE NO		NOT HEX	
2B	8.1.4.6		CMP A #46		A ≤ 46	
2D	2.F.F.6		BLE YES		THEN ITS HEX	
2F	0.0	NO	SEC		IF NOT SET CARRY	
30	3.9		RTS		AND RETURN	
31	8.5.3.0	BINARY	BIT A #30		IS ACC A A LETTER?	
33	2.7.0.C		BEQ LETT			
35	4.8	R10A	ASL A		ITS A NUMBER SO	
36	4.8		ASL A		GET RID OF MOST	
37	4.8		ASL A		SIGNIFICANT 4 BITS.	
38	4.8		ASL A			
39	C.5.3.0		BIT B #30		IS ACC B A LETTER?	
3B	2.7.0.8		BEQ LETTER			
3D	C.4.0.F	R10B	AND B #0F		DESTROY MS 4 BITS.	
3F	1.B		ABA		FORM BINARY CHAR.	
40	3.9		RTS		RETURN	
41	8.B.0.9	LETT	ADD A #09		MAKE ACC A BINARY	
43	2.0.F.0		BRA R10A		AS BEFORE	
45	C.B.0.9	LETTER	ADD B #09		MAKE ACC B BINARY	
47	2.0.F.4		BRA R10B		AS BEFORE	
49	8.0.B.5	ZIN	BSR R0		GET A CHARACTER	
4B	8.0.0.0		BSR VHEX		IS IT 0 → 9 or A → F?	
4D	2.5.1.0		BCS ?		NO - ITS RUBBISH SO ?	
4F	B.7.F.0.F.6		STA A Q		YES - ITS OK SO SAVE IT	
52	8.0.A.C		BSR R0		GET SECOND CHARACTER	
54	8.0.C.7		BSR VHEX		IS IT HEX	
56	2.5.0.7		BCS ?		NO - PRINT A'?	
58	1.6		TAB		YES - PUT IT INTO ACC B	



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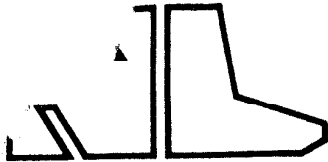
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PROGRAM MINIMON		VERSION		AUTHOR ACH	DATE DEC '78	PAGE 30/14
ADDRESS	MACHINE CODE	LABEL	OPERATOR & OPERAND		COMMENTS	
FC59	86F0F6		LDA A Q		RETRIEVE FIRST CHARACTER	
5C	8003		BSR BINARY		CONVERT ACCA,B TO BINARY	
5E	39		RTS		RETURN	
5F	863F	?	LOA A #'?		LOAD ACCA WITH ASCII '?'	
61	80AC		BSR PR		PRINT IT	
63	20E4		BRA 2IN		GO BACK TO START	
65	80FD40	RDX	SSR PR		PRINT A SPACE ()	
68	800F		BSR 2IN		INPUT TWO HEX CHARACTERS	
6A	87F0F3		STA A Z		SAVE BYTE IN MS. Z	
6D	800A		BSR 2IN		GET LS. HEX CHARACTERS	
6F	87F0F4		STA A Z+1		STORE IN LS. Z	
72	FEF0F3		LDX Z		X ← Z	
75	39		RTS		RETURN	
76	30	STRING	TSX		POINT X AT LOCATION ON	
77	EE00		LDX X		STACK WHERE RETURN	
79	09		DEX		ADD. IS HELD. LOAD X WITH	
7A	08	AGAIN	INX		RETURN ADD.	
7B	A600		LOA A X		GET DATA TO BE PRINTED	
7D	81FF		CMP A #FF		LAST BYTE?	
7F	2704		BEQ END		YES - GO TO THE END	
81	808C		BSR PR		NO - PRINT BYTE	
83	20F5		BRA AGAIN			
85	31	END	INS		CLEAN UP STACK	
86	31		INS			
87	6E01		JMD X,01		RETURN TO CALLER	
89	80EB	GETADD.	BSR STRING		PRINT:	
8B	2053FF				(SPACE), S	
8E	8005		BSR RDX		INPUT 4 DIGIT HEX ADDRESS	
90	FFF0EF		STX START		STORE IT IN START	
93	80E1		BSR STRING		PRINT:	
95	2046FF				(SPACE), F	
98	80CB		BSR RDX		INPUT 4 DIGIT HEX ADDRESS	
9A	FFF0F1		STX STOP		STORE IT IN STOP	
9D	39		RTS		RETURN TO CALLER	
9E	16	ASCII	TAB		ACCB ← ACCA	



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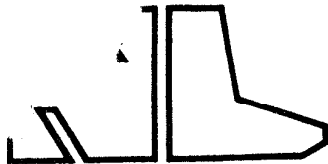
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PROGRAM <i>MINIMON</i>		VERSION	AUTHOR <i>ACH</i>	DATE <i>DEC '78</i>	PAGE <i>4</i> OF <i>14</i>
ADDRESS	MACHINE CODE	LABEL	OPERATOR & OPERAND	COMMENTS	
<i>FC9F</i>	<i>84F0</i>		<i>AND A #F0</i>		
<i>A1</i>	<i>44</i>		<i>LSR A</i>	<i>SHIFT MOST SIGNIFICANT</i>	
<i>A2</i>	<i>44</i>		<i>LSR A</i>	<i>4 BITS DOWN.</i>	
<i>A3</i>	<i>44</i>		<i>LSR A</i>		
<i>A4</i>	<i>44</i>		<i>LSR A</i>		
<i>A5</i>	<i>8109</i>		<i>CMP A #09</i>	<i>LETTER OR NUMBER?(MS)</i>	
<i>A7</i>	<i>220B</i>		<i>BHI SUBT</i>		
<i>A9</i>	<i>8B30</i>		<i>ADD A #30</i>	<i>MAKE IT A NUMBER(ASCII)</i>	
<i>AB</i>	<i>C40F</i>	<i>AND</i>	<i>AND B #0F</i>	<i>GET RID OF MS 4 BITS</i>	
<i>AD</i>	<i>C109</i>		<i>CMP B #09</i>	<i>LETTER OR NUMBER?(LS)</i>	
<i>AF</i>	<i>2207</i>		<i>BHI ADD</i>		
<i>B1</i>	<i>CB30</i>		<i>ADD B #30</i>	<i>MAKE IT AN ASCII NUMBER</i>	
<i>B3</i>	<i>39</i>		<i>RTS</i>	<i>FINNISHED</i>	
<i>B4</i>	<i>8B37</i>	<i>SUBT</i>	<i>ADD A #37</i>	<i>MAKE IT AN ASCII LETTER</i>	
<i>B6</i>	<i>20F3</i>		<i>BRA AND</i>	<i>THEN DO LS CHARACTER</i>	
<i>B8</i>	<i>CB37</i>	<i>ADD</i>	<i>ADD B #37</i>	<i>MAKE IT AN ASCII LETTER</i>	
<i>BA</i>	<i>39</i>		<i>RTS</i>	<i>RETURN</i>	
<i>BB</i>	<i>FFFD,ED</i>	<i>PRX</i>	<i>STX TEMPX</i>	<i>STORE THE INDEX REG.</i>	
<i>BE</i>	<i>66FD,ED</i>		<i>LDA A TEMPX_H</i>	<i>ACCA ← X_H</i>	
<i>C1</i>	<i>8006</i>		<i>BSR ZOUT</i>	<i>PRINT ACCA IN HEX</i>	
<i>C3</i>	<i>B6FD,EE</i>		<i>LDA A TEMPX_L</i>	<i>ACCA ← X_L</i>	
<i>C6</i>	<i>8001</i>		<i>BSR ZOUT</i>	<i>PRINT ACCA IN HEX</i>	
<i>C8</i>	<i>39</i>		<i>RTS</i>	<i>RETURN</i>	
<i>C9</i>	<i>8003</i>	<i>ZOUT</i>	<i>BSR ASCII</i>	<i>ACCA(BINARY) → ACCA,B(ASCII)</i>	
<i>CB</i>	<i>F7FD,EC</i>		<i>STA B M</i>	<i>SAVE ACCB</i>	
<i>CE</i>	<i>BD,FC,0F</i>		<i>JSR PR</i>	<i>PRINT ACCA</i>	
<i>D1</i>	<i>BD,FC,0F</i>		<i>LDA A M</i>	<i>RETRIEVE ACCB</i>	
<i>D4</i>	<i>BD,FC,0F</i>		<i>JSR PR</i>	<i>PRINT IT</i>	
<i>D7</i>	<i>39</i>		<i>RTS</i>	<i>RETURN</i>	
<i>D8</i>	<i>BD,FC,89</i>	<i>PUNCH</i>	<i>JSR GETADD</i>	<i>GET BOUNDARY ADDRESSES</i>	
<i>DB</i>	<i>CE0000</i>		<i>LOX #0000</i>	<i>WAIT APPROXIMATELY</i>	
<i>DE</i>	<i>80BE</i>	<i>LOOP</i>	<i>BSR ASCII</i>	<i>5 SECS.</i>	
<i>E0</i>	<i>08</i>		<i>INX</i>		
<i>E1</i>	<i>26FB</i>		<i>BNE LOOP</i>		
<i>E3</i>	<i>8014</i>		<i>BSR BEGIN</i>	<i>START OF RECORD</i>	



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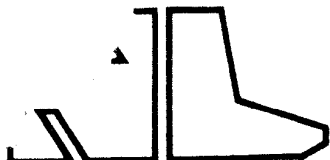
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HEXADECIMAL CODING FORM

PROGRAM MINIMON		VERSION		AUTHOR ACH		DATE DEC '78		PAGE 5 OF 14	
ADDRESS	MACHINE CODE	LABEL	OPERATOR & OPERAND		COMMENTS				
FC	ES	FE	F0	EF		LDX	START	POINT X AT DATA TO	
	E8	09				DEX		BE PUNCHED.	
	E9	08			WRITE	INX			
	EA	A6	00			LDA	A X	GET BYTE	
	EC	BD	FC	0F		JSR	PR	PUNCH IT *	
	EF	BC	F0	F1		CPX	STOP	LAST BYTE?	
	F2	26	F5			BNE	WRITE	NO - THEN REPEAT	
	F4	8D	35			BSR	R0B	YES - WAIT FOR INPUT	
	F6	7E	FF	8F		JMP	START	FROM TERMINAL; RETURN	
	F9	86	53		BEGIN	LDA	A #53	PUNCH TWO 'S' TO	
	FB	BD	FC	0F		JSR	PR	INDECTE START OF	
	FE	BD	FC	0F		JSR	PR	DATA BLOCK ON TAPE	
FD	01	86	11		BIN	LDA	A #11	SET UP ACIA(a) FOR:	
	03	67	F4	01		STA	A CTRLA	8 BITS + 2 STOP BITS +	
	06	39				RTS		÷ 16 CLOCK.	
	07	BD	FC	89	LOAD	JSR	GETADD	GET BOUNDRY ADDRESSES	
	0A	8D	1F		NEXT	BSR	R0B	BEGIN TO READ DATA	
	0C	81	53			CMP	A #53	BUT DONT STORE IT	
	0E	26	FA			BNE	NEXT	UNTILL TWO 'S' (ASCII)	
	10	8D	19			BSR	R0B	HAVE BEEN READ.	
	12	81	53			CMP	A #53		
	14	26	F4			BNE	NEXT		
	16	8D	E9			BSR	BIN	THEN SET UP ACIA(a)	
	18	FE	F0	EF		LDX	START	POINT X AT PROGRAM	
	1B	09				DEX		AREA	
	1C	08			READ	INX			
	1D	8D	04			BSR	R0B	INPUT BYTE	
	1F	AA	70	01		STA	A X	STORE IT	
	21	BC	F0	F1		CPX	STOP	LAST BYTE?	
	24	26	F6			BNE	READ	NO - CARRY ON	
	26	7E	FF	8F		JMP	START	YES - GOTO START OF MINIMON	
	29								
	2B	86	F4	01	R0B	LDA	A CTRLA	ACIA(a) RECIVED A CHAR.	
	2E	85	01			BIT	A #01		
	30	27	F9			BEQ	R0B	NO - TRY AGAIN	



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HEXADECIMAL CODING FORM

PROGRAM MINIMON		VERSION		AUTHOR ACH	DATE DEC '78	PAGE 6 OF 14
ADDRESS	MACHINE CODE	LABEL	OPERATOR & OPERAND		COMMENTS	
F032	66F400		LOA A DATAA		YES- GET IT	
35	39		RTS		AND RETURN	
36	8D15	REG.PRINT	BSR PR0		PRINT A SPACE	
38	FEF0FD		LDX PSTACK		POINT X REG. AT USERS STACK	
3B	8D14		BSR PR2		PRINT CC	
3D	8D1A		BSR PR2		" B	
3F	8D18		BSR PR2		" A	
41	8D10		BSR PR4		" X	
43	8D0E		BSR PR4		" PC	
45	CEF0FD		LDX #PSTACK		POINT X AT 'STACK POINTER'	
48	8D0A		BSR 4PR		PRINT SP	
4A	7EFF8F		JMP START		RETURN TO MINIMON	
4D	8620	PR0	LOA A #20		LOAD ACCA WITH AN ASCII	
4F	60FC0F		JSR PR		SPACE AND PRINT IT	
52	39		RTS		RETURN	
53	08	PR4	INX		X ← X + 1	
54	A600	4PR	LOA A X		GET DATA(X)	
56	60FCC9		JSR 2OUT		PRINT IT	
59	08	PR2	INX		X ← X + 1	
5A	A600		LOA A X		GET DATA(X)	
5C	60FCC9		JSR 2OUT		PRINT IT	
5F	8DEC		BSR PR0		PRINT A SPACE	
61	39		RTS		RETURN	
62	80FC89	BLOCKMOVE	JSR GETADD		GET BLOCK LIMITS	
65	60FC76		JSR STRING		PRINT:	
68	20544F				V TO	
6B	FF					
6C	60FC65		JSR RDX		GET NEW START ADDRESS	
6F	FFFD0EA		STX NEW		STORE IT	
72	FFFD0E6		STX X		ALSO PUT IT IN X	
75	FFFD0EF		LDX START		LOAD X REG WITH START	
78	FFFD0E8		STX Y		TRANSFER IT TO Y	
7B	8059		BSR SUB		DO: X - Y	
7D	2A23		BSPL DOWN		IF RESULT IS +VE THEN ITS	
7F	FFFD0F1		LDX STOP		DOWN.	



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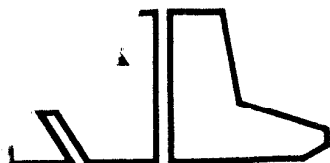
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PROGRAM <i>MINIMON</i>		VERSION		AUTHOR <i>ACH</i>		DATE <i>DEC '78</i>	PAGE <i>7</i> OF <i>14</i>
ADDRESS	MACHINE CODE	LABEL	OPERATOR & OPERAND		COMMENTS		
<i>F082</i>	<i>08</i>		<i>INX</i>		<i>ADD 1 to STOP</i>		
<i>83</i>	<i>FF F0 F1</i>		<i>STX STOP</i>		<i>AND STORE IT</i>		
<i>86</i>	<i>FE F0 EF</i>	<i>DO</i>	<i>LDX START</i>		<i>GET START</i>		
<i>89</i>	<i>BC F0 F1</i>		<i>CPX STOP</i>		<i>COMPARE IT WITH STOP</i>		
<i>8C</i>	<i>26 03</i>		<i>BNE CARRYON</i>		<i>FINNISHED ?</i>		
<i>8E</i>	<i>7E FF 8F</i>	<i>END</i>	<i>JMP START</i>		<i>YES - RETURN TO MINIMON</i>		
<i>91</i>	<i>A6 00</i>	<i>CARRYON</i>	<i>LDA A X</i>		<i>NO - GET DATA</i>		
<i>93</i>	<i>08</i>		<i>INX</i>		<i>INCREMENT START</i>		
<i>94</i>	<i>FF F0 EF</i>		<i>STX START</i>		<i>STORE IT</i>		
<i>97</i>	<i>FE F0 EA</i>		<i>LDX NEW</i>		<i>LOAD X REG WITH NEW</i>		
<i>9A</i>	<i>A7 00</i>		<i>STA A X</i>		<i>STORE DATA</i>		
<i>9C</i>	<i>08</i>		<i>INX</i>		<i>INCREMENT NEW</i>		
<i>9D</i>	<i>FF F0 EA</i>		<i>STX NEW</i>		<i>STORE IT</i>		
<i>AD</i>	<i>20 E4</i>		<i>BRA DO</i>		<i>ROUND AGAIN</i>		
<i>A2</i>	<i>FE F0 F1</i>	<i>DOWN</i>	<i>LDX STOP</i>		<i>LOAD X REG WITH STOP</i>		
<i>A5</i>	<i>FF F0 E6</i>		<i>STX X</i>		<i>PUT IT IN X</i>		
<i>A8</i>	<i>8D 2C</i>		<i>BSR SUB</i>		<i>ACCA, B ← X - Y</i>		
<i>AA</i>	<i>BB F0 E6</i>		<i>ADD A NEWL</i>				
<i>AD</i>	<i>FB F0 EA</i>		<i>ADD B NEWH</i>		<i>NEW ← NEW + ACCA, B</i>		
<i>AB</i>	<i>B7 F0 E6</i>		<i>STA A NEWL</i>				
<i>B3</i>	<i>F7 F0 EA</i>		<i>STA B NEWH</i>				
<i>B6</i>	<i>FE F0 EF</i>		<i>LDX START</i>		<i>GET START</i>		
<i>B9</i>	<i>09</i>		<i>DEX</i>		<i>DECREMENT IT</i>		
<i>BA</i>	<i>FF F0 EF</i>		<i>STX START</i>		<i>SAVE START</i>		
<i>BD</i>	<i>FE F0 F1</i>	<i>DO IT</i>	<i>LDX STOP</i>		<i>LOAD X REG WITH STOP</i>		
<i>CD</i>	<i>BC F0 EF</i>		<i>CPX START</i>		<i>COMPARE IT WITH START</i>		
<i>C3</i>	<i>27 C9</i>		<i>BEQ END</i>		<i>IF EQUAL THEN END</i>		
<i>C5</i>	<i>A6 00</i>		<i>LDA A X</i>		<i>ACC A ← X STOP</i>		
<i>C7</i>	<i>09</i>		<i>DEX</i>		<i>STOP ← STOP - 1</i>		
<i>C8</i>	<i>FF F0 F1</i>		<i>STX STOP</i>				
<i>CB</i>	<i>FE F0 EA</i>		<i>LDX NEW</i>		<i>LOAD X REG WITH NEW</i>		
<i>CE</i>	<i>A7 00</i>		<i>STA A X</i>		<i>STORE DATA</i>		
<i>D0</i>	<i>09</i>		<i>DEX</i>		<i>NEW ← NEW - 1</i>		
<i>D1</i>	<i>FF F0 EA</i>		<i>STX NEW</i>		<i>STORE NEW</i>		
<i>D4</i>	<i>20 E7</i>		<i>BRA DO IT</i>		<i>AND AGAIN</i>		



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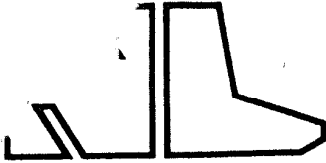
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PROGRAM <i>MINIMON</i>		VERSION	AUTHOR <i>ACH</i>	DATE <i>DEC'78</i>	PAGE <i>8 of 14</i>
ADDRESS	MACHINE CODE	LABEL	OPERATOR & OPERAND	COMMENTS	
<i>F0,06</i>	<i>CE,F0E6</i>	<i>SUB</i>	<i>LDX #X</i>	<i>POINT XREG AT DATA-</i>	
<i>D9</i>	<i>A6,01</i>		<i>LDA A XL</i>	<i>AREA. ACCA ← XL</i>	
<i>06</i>	<i>EG,00</i>		<i>LDA B XH</i>	<i>ACCB ← XH</i>	
<i>00</i>	<i>A0,03</i>		<i>SUB A YL</i>	<i>ACCA ← ACCA - YL</i>	
<i>0F</i>	<i>E2,02</i>		<i>SBC B YH</i>	<i>ACCB ← ACCB - YH - C</i>	
<i>E1</i>	<i>39</i>		<i>RTS</i>	<i>RETURN</i>	
<i>E2</i>	<i>8D,4F</i>	<i>Z</i>	<i>BSR DP</i>	<i>PRINT DP FOR PC</i>	
<i>E4</i>	<i>BD,FC,49</i>		<i>JSR ZIN</i>	<i>GET PC ADDRESS</i>	
<i>E7</i>	<i>67,F0E5</i>		<i>STA A P</i>	<i>SAVE IT</i>	
<i>EA</i>	<i>8D,4F</i>		<i>BSR DP</i>	<i>PRINT DP FOR DESTINATION</i>	
<i>EC</i>	<i>BD,FC,49</i>		<i>JSR ZIN</i>	<i>GET D ADDRESS</i>	
<i>EF</i>	<i>6D,F0E5</i>		<i>SUB A P</i>	<i>D ← D - P</i>	
<i>F2</i>	<i>23,17</i>		<i>BLS BACK</i>	<i>BRANCHING BACK OR FORWARD?</i>	
<i>F4</i>	<i>80,02</i>	<i>FORWARD</i>	<i>SUB A #02</i>	<i>FORWARD: D ← D - 2</i>	
<i>F6</i>	<i>2B,17</i>		<i>BMI ERROR</i>	<i>IF D > 127 THEN ERROR!</i>	
<i>F8</i>	<i>67,F0E5</i>	<i>PRINT</i>	<i>STA A P</i>	<i>SAVE D</i>	
<i>F0</i>	<i>BD,FC,76</i>		<i>JSR STRING</i>	<i>PRINT:</i>	
<i>FE</i>	<i>20,52,3D</i>			<i>DR =</i>	
<i>FE,01</i>	<i>FF</i>				
<i>02</i>	<i>B6,F0E5</i>		<i>LDA A P</i>	<i>RETRIEVE D</i>	
<i>05</i>	<i>BD,FC,49</i>		<i>JSR ZOUT</i>	<i>PRINT IT</i>	
<i>08</i>	<i>7E,FF,8F</i>	<i>END</i>	<i>JMP START</i>	<i>RETURN TO MINIMON</i>	
<i>00</i>	<i>80,02</i>	<i>BACK</i>	<i>SUB A #02</i>	<i>BACKWARDS: D ← D - 2</i>	
<i>00</i>	<i>2B,E9</i>		<i>BMI PRINT</i>	<i>IF D ≥ 128 THEN OK SO PRINT</i>	
<i>0F</i>	<i>BD,FC,76</i>	<i>ERROR</i>	<i>JSR STRING</i>	<i>PRINT:</i>	
<i>12</i>	<i>20,52,41</i>				
<i>15</i>	<i>4E,47,45</i>			<i>RANGE!</i>	
<i>18</i>	<i>21,FF</i>				
<i>1A</i>	<i>20,EC</i>		<i>BRA END</i>	<i>FINNISH</i>	
<i>1C</i>	<i>8D,15</i>	<i>X</i>	<i>BSR DP</i>	<i>PRINT P FOR PL</i>	
<i>1E</i>	<i>BD,FC,65</i>		<i>BSR RDX</i>	<i>GET P (16 BIT)</i>	
<i>21</i>	<i>FF,F0E8</i>		<i>STX Y</i>	<i>STORE P IN Y</i>	
<i>24</i>	<i>8D,15</i>		<i>BSR DP</i>	<i>PRINT D FOR DESTINATION</i>	
<i>26</i>	<i>BD,FC,65</i>		<i>JSR RDX</i>	<i>GET D (16 BIT)</i>	
<i>29</i>	<i>FF,F0E6</i>		<i>STX X</i>	<i>STORE D IN X</i>	



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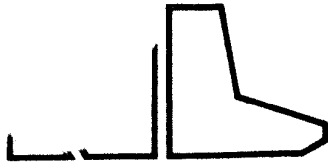
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PROGRAM <i>MINIMON</i>		VERSION		AUTHOR <i>ACH</i>	DATE <i>DEC '78</i>	PAGE <i>9 of 4</i>
ADDRESS	MACHINE CODE	LABEL	OPERATOR & OPERAND		COMMENTS	
<i>FE2C</i>	<i>60 FD 06</i>		<i>JSR SUB</i>		<i>ACC ← X - Y</i>	
<i>2F</i>	<i>2B DA</i>		<i>BMI BACK</i>		<i>BACK OR FORWARD?</i>	
<i>31</i>	<i>20 C1</i>		<i>BRA FORWARD</i>			
<i>33</i>	<i>60 FC 76</i>	<i>VP</i>	<i>JSR STRING</i>		<i>PRINT:</i>	
<i>36</i>	<i>20 50 20</i>				<i>VP</i>	
<i>39</i>	<i>FF</i>					
<i>3A</i>	<i>39</i>		<i>RTS</i>		<i>RETURN</i>	
<i>3B</i>	<i>60 FC 76</i>	<i>VD</i>	<i>JSR STRING</i>		<i>PRINT:</i>	
<i>3E</i>	<i>20 44 20</i>				<i>VD</i>	
<i>41</i>	<i>FF</i>					
<i>42</i>	<i>39</i>		<i>RTS</i>		<i>RETURN</i>	
<i>43</i>	<i>60 FC 65</i>	<i>MODIFY</i>	<i>JSR K0X</i>		<i>GET ADDRESS</i>	
<i>46</i>	<i>60 FD 40</i>	<i>STRT</i>	<i>JSR PR</i>		<i>PRINT A SPACE</i>	
<i>49</i>	<i>7F FD FS</i>	<i>STRT</i>	<i>CLR R</i>		<i>R ← 0</i>	
<i>4C</i>	<i>A6 00</i>	<i>PR</i>	<i>LDA A X</i>		<i>GET DATA</i>	
<i>4E</i>	<i>60 FC C9</i>		<i>JSR ZOUT</i>		<i>PRINT IT</i>	
<i>51</i>	<i>60 FC 00</i>	<i>WAIT</i>	<i>JSR K0</i>		<i>GET COMMAND</i>	
<i>54</i>	<i>81 20</i>	<i>WHAT</i>	<i>CMP A #20</i>		<i>IS IT A SPACE?</i>	
<i>56</i>	<i>26 0C</i>		<i>BNE /?</i>		<i>NO - WHAT THEN</i>	
<i>58</i>	<i>08</i>	<i>CARRY ON</i>	<i>INX</i>		<i>YES - INC X</i>	
<i>59</i>	<i>66 FD FS</i>		<i>LDA A R</i>		<i>ACC A ← R</i>	
<i>5C</i>	<i>81 16</i>		<i>CMP A #16</i>		<i>AT END OF THE LINE?</i>	
<i>5E</i>	<i>2F EC</i>		<i>BLE PR</i>		<i>NO - CARRY ON</i>	
<i>60</i>	<i>80 35</i>	<i>NEW</i>	<i>JSR NEWLINE</i>		<i>YES - PRINT A NEW LINE</i>	
<i>62</i>	<i>20 E2</i>		<i>BRA STRT</i>		<i>CARRY ON</i>	
<i>64</i>	<i>81 2F</i>	<i>/?</i>	<i>CMP A #2F</i>		<i>IS IT A SLASH?</i>	
<i>66</i>	<i>26 03</i>		<i>BNE CR?</i>		<i>NO - WHAT THEN?</i>	
<i>68</i>	<i>09</i>		<i>DEX</i>		<i>YES - X ← X - 1</i>	
<i>69</i>	<i>20 FS</i>		<i>BRA NEW</i>		<i>CARRY ON</i>	
<i>6B</i>	<i>81 00</i>	<i>CR?</i>	<i>CMP A #00</i>		<i>IS IT A CARRIAGE RETURN</i>	
<i>6D</i>	<i>26 03</i>		<i>BNE DATA?</i>		<i>NO - WHAT THEN?</i>	
<i>6F</i>	<i>08</i>		<i>INX</i>		<i>YES - X ← X + 1</i>	
<i>70</i>	<i>20 EE</i>		<i>BRA NEW</i>		<i>CARRY ON</i>	
<i>72</i>	<i>60 FC 10</i>	<i>DATA?</i>	<i>JSR VHEX</i>		<i>IS IT A VALID HEX CHAR.</i>	
<i>75</i>	<i>25 19</i>		<i>BCS ?</i>		<i>NO - PRINT ?</i>	



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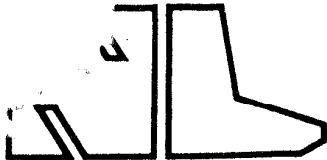
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PROGRAM <i>MINIMON</i>		VERSION		AUTHOR <i>ACH</i>	DATE <i>DEC '78</i>	PAGE/00F/14
ADDRESS	MACHINE CODE	LABEL	OPERATOR & OPERAND		COMMENTS	
<i>FE 77</i>	<i>B7 F0 F6</i>		<i>STA AQ</i>		<i>YES - SO SAVE IT</i>	
<i>7A</i>	<i>BD FC 00</i>		<i>JSR R0</i>		<i>GET SECOND CHARACTER</i>	
<i>7D</i>	<i>BD FC 10</i>		<i>JSR VHEX</i>		<i>IS IT HEX ?</i>	
<i>80</i>	<i>25 02</i>		<i>KCS WHAT</i>		<i>NO - THEN WHAT IS IT</i>	
<i>82</i>	<i>16</i>		<i>TAB</i>		<i>YES - ARRANGE FOR</i>	
<i>83</i>	<i>B6 F0 F6</i>		<i>LDA AQ</i>		<i>CONVERSION TO BINARY</i>	
<i>86</i>	<i>BD FC 31</i>		<i>JSR BINARY</i>		<i>CONVERT</i>	
<i>89</i>	<i>A7 00</i>		<i>STA AX</i>		<i>STORE IT AT THE RIGHT</i>	
<i>8B</i>	<i>BD FD 40</i>		<i>JSR PR</i>		<i>PLACE, PRINT A SPACE</i>	
<i>8E</i>	<i>20 C8</i>		<i>BRA CARRYON</i>		<i>CARRY ON</i>	
<i>90</i>	<i>86 3F</i>	<i>?</i>	<i>LDA A #3F</i>		<i>ACCA ← ASCII ?</i>	
<i>92</i>	<i>BD FC 0F</i>		<i>JSR PR</i>		<i>PRINT ACCA</i>	
<i>95</i>	<i>20 BA</i>		<i>BRA WAIT</i>		<i>CARRY ON,</i>	
<i>97</i>	<i>FF F0 E3</i>	<i>NEWLINE</i>	<i>STX SAVE</i>		<i>SAVE X REG</i>	
<i>9A</i>	<i>BD FC 76</i>		<i>JSR STRING</i>		<i>PRINT:</i>	
<i>9D</i>	<i>00 0A 00</i>				<i>CR, LF, NULL</i>	
<i>A0</i>	<i>FF</i>					
<i>A1</i>	<i>FE F0 E3</i>		<i>LDX SAVE</i>		<i>RESTORE X REG</i>	
<i>A4</i>	<i>BD FC BB</i>		<i>JSR PRX</i>		<i>PRINT IT</i>	
<i>A7</i>	<i>39</i>		<i>RTS</i>		<i>RETURN</i>	
<i>AB</i>	<i>BD FC 65</i>	<i>ALTER</i>	<i>JSR R0X</i>		<i>GET START ADDRESS</i>	
<i>AB</i>	<i>7F F0 F5</i>	<i>NEW</i>	<i>CLR R</i>		<i>R ← 0</i>	
<i>AE</i>	<i>80 E7</i>		<i>BSR NEWLINE</i>		<i>PRINT 'NEWLINE'</i>	
<i>B0</i>	<i>BD F0 40</i>		<i>JSR PR</i>		<i>PRINT SPACE</i>	
<i>B2</i>	<i>BD FC 49</i>	<i>GET</i>	<i>JSR ZIN</i>		<i>GET DATA</i>	
<i>B6</i>	<i>A7 00</i>		<i>STA AX</i>		<i>STORE IT</i>	
<i>B8</i>	<i>08</i>		<i>INX</i>		<i>X ← X + 1</i>	
<i>B9</i>	<i>B6 F0 F5</i>		<i>LDA AR</i>		<i>ACCA ← R</i>	
<i>BC</i>	<i>81 26</i>		<i>CMP A #26</i>		<i>AT END OF LINE ?</i>	
<i>BE</i>	<i>2F F3</i>		<i>BLE GET</i>		<i>NO - CARRY ON</i>	
<i>C0</i>	<i>20 E9</i>		<i>BRA NEW</i>		<i>YES - PRINT A NEW LINE</i>	
<i>C2</i>	<i>BD FC 65</i>	<i>GO</i>	<i>JSR R0X</i>		<i>GET PROGRAM START ADD.</i>	
<i>C5</i>	<i>BD FC 00</i>		<i>JSR R0</i>		<i>RIGHT ADDRESS ?</i>	
<i>C8</i>	<i>B6 F0 F4</i>		<i>LDA A ZL</i>		<i>ACC ← ADDRESS</i>	
<i>CA</i>	<i>F6 F0 F3</i>		<i>LDA B ZH</i>			



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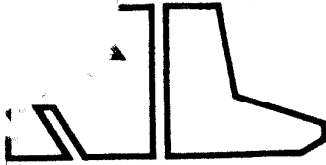
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HEXADECIMAL CODING FORM

PROGRAM <i>MINIMON</i>		VERSION	AUTHOR <i>ACH</i>	DATE <i>DEC '78</i>	PAGE <i>11</i> OF <i>14</i>
ADDRESS	MACHINE CODE	LABEL	OPERATOR & OPERAND	COMMENTS	
<i>FECE</i>	<i>FEF0FD</i>	<i>GOTO</i>	<i>LDX PSTACK</i>	<i>LOAD XREG WITH PSTACK</i>	
<i>.D1</i>	<i>A707</i>		<i>STA A X,07</i>	<i>PUT NEW RETURN</i>	
<i>.D3</i>	<i>E706</i>		<i>STA B X,06</i>	<i>ADDRESS ON STACK</i>	
<i>.D5</i>	<i>BEF0FD</i>		<i>LDS PSTACK</i>	<i>SET UP PROGRAMS STACK</i>	
<i>.D8</i>	<i>36</i>		<i>RTI</i>	<i>GO TO USER PROGRAM</i>	
<i>.D9</i>	<i>BD FC0D</i>	<i>CONTINUE</i>	<i>JSR RD</i>	<i>1 OF 2</i>	
<i>.DC</i>	<i>8131</i>		<i>CMP A #31</i>	<i>IS IT A 1?</i>	
<i>.DE</i>	<i>2610</i>		<i>BNE 2?</i>	<i>NO- 2 THEN</i>	
<i>.E0</i>	<i>FEF0F7</i>		<i>LDX A BYTE</i>	<i>YES- GET ADDRESS OF BR. PT.</i>	
<i>.E3</i>	<i>B6F0FB</i>		<i>LDA A BYTE</i>	<i>GET SAVED OPP. CODE</i>	
<i>.E6</i>	<i>A700</i>		<i>STA A X</i>	<i>PUT IT BACK IN THE PROG.</i>	
<i>.E8</i>	<i>B6F0FB</i>		<i>LDA A A BYTE</i>	<i>GET ADDRESS AT WHICH</i>	
<i>.E0</i>	<i>F6F0F7</i>		<i>LDA B A BYTE</i>	<i>TO RE-ENTER PROGRAM</i>	
<i>.EE</i>	<i>20DE</i>		<i>BRA GOTO</i>	<i>GO TO USERS PROGRAM</i>	
<i>.FD</i>	<i>8132</i>	<i>2?</i>	<i>CMP A #32</i>	<i>IS IT A 2?</i>	
<i>.F2</i>	<i>2703</i>		<i>BEQ 2</i>	<i>YES-OK</i>	
<i>.F4</i>	<i>7EFF8F</i>		<i>SWP START</i>	<i>NO - RETURN TO MINIMON</i>	
<i>.F7</i>	<i>FEF0F9</i>	<i>2</i>	<i>LDX A BYTE</i>	<i>GET ADDRESS OF BR. PT 2</i>	
<i>.FA</i>	<i>B6F0FC</i>		<i>LDA A BYTE</i>	<i>GET OLD OPP. CODE</i>	
<i>.F0</i>	<i>A700</i>		<i>STA A X</i>	<i>PUT IT BACK</i>	
<i>.FF</i>	<i>B6F0FA</i>		<i>LDA A A BYTE</i>	<i>LOAD ADDRESS TO START</i>	
<i>FF02</i>	<i>F6F0F9</i>		<i>LDA B A BYTE</i>	<i>RUNNING PROGRAM AT</i>	
<i>.05</i>	<i>20C7</i>		<i>BRA GOTO</i>	<i>RUN USER PROGRAM.</i>	
<i>.07</i>	<i>30FC89</i>	<i>DUMP</i>	<i>JSR GETADD</i>	<i>GET COUNDRY ADDRESSES</i>	
<i>.0A</i>	<i>FEF0EF</i>		<i>LDX START</i>		
<i>.0D</i>	<i>B0FE97</i>	<i>NEW</i>	<i>JSR NEWLINE</i>	<i>PRINT A NEW LINE</i>	
<i>.10</i>	<i>8608</i>		<i>LDA A #08</i>		
<i>.12</i>	<i>B7F0F6</i>		<i>STA A Q</i>	<i>Q ← 8</i>	
<i>.15</i>	<i>09</i>		<i>DEX</i>		
<i>.16</i>	<i>08</i>	<i>NEXT</i>	<i>INX</i>		
<i>.17</i>	<i>B0FD40</i>		<i>JSR PRG</i>	<i>PRINT A SPACE</i>	
<i>.1A</i>	<i>A600</i>		<i>LDA A X</i>	<i>GET DATA</i>	
<i>.1C</i>	<i>B0FC99</i>		<i>JSR ZOUT</i>	<i>PRINT IT</i>	
<i>.1F</i>	<i>BCF0F1</i>		<i>CPX STOP</i>	<i>FINNISHED ?</i>	
<i>.22</i>	<i>2606</i>		<i>BNE CARRYON</i>	<i>NO-CARRY ON</i>	



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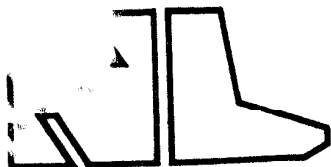
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HEXADECIMAL CODING FORM

PROGRAM MINIMON		VERSION	AUTHOR ACH	DATE DEC '79	PAGE 2 OF 14
ADDRESS	MACHINE CODE	LABEL	OPERATOR & OPERAND	COMMENTS	
FF24	B6F400	END	LDA A DATAA		
27	7EFF8F		JMP START	RETURN TO MINIMON	
2A	B6F401	CARRY ON	LDA A CTRLA		
2D	8501		BIT A #01	STOP DUMP?	
2F	26F3		BNE END	YES-OK	
31	7AF0F6		DEC C	NO-CARRY ON, NEW LINE?	
34	26E0		BNE NEXT	NO-OK	
36	08		INX	YES-PRINT NEXT BYTE	
37	20D4		BRA NEW	ON A NEW LINE	
39	BDFC00	BR.PT.SET	JSR RD	IS IT A 1 OR A 2?	
3C	8131		CMP A #31		
3E	2612		BNE 2?		
40	BDFC65		JSR RDX	ITS A1 SO GET THE ADD.	
43	A600		LDA A X	GET THE OPP CODE	
45	B7F0FB		STA A BYTE1	SAVE IT	
48	FF0F7		STX A BYTE1	ALSO SAVE THE ADDRESS	
4D	863F	SAME	LDA A #3F	SET THE	
4D	A700		STA A X	BREAK POINT	
4F	7EFF8F	END	JMP START	FINNISHED	
52	8132	2?	CMP A #32	IS IT A 2?	
54	26F9		BNE END	NO-THEN ITS RUBBISH	
56	BDFC65		JSR RDX	YES-GET ADDRESS	
59	A600		LDA A X	GET OPP.CODE	
5B	B7F0FC		STA A BYTE2	SAVE IT	
5E	FF0F9		STX A BYTE2	SAVE ADDRESS	
61	20E8		BRA SAME	SET THE BREAK POINT	
63	BDFC76	HEADER	JSR STRING	PRINT THE FOLLOWING	
66	204343			ASCII STRING WHICH	
69	204220			IS TERMINATED WITH	
6C	204120			HEX FF:	
6F	202058				
72	202020			CC B A X PC SP	
75	205043				
78	202020				
7D	5350FF				



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ADDRESS	MACHINE CODE	LABEL	OPERATOR & OPERAND	COMMENTS	
<u>FF7E</u>	<u>7E FF 8F</u>		<u>JMP START</u>	<u>RETURN TO MINIMON</u>	
<u>81</u>	<u>8E F0 00</u>	<u>RESET</u>	<u>LDS #STACK</u>	<u>SET UP STACK FOR MINIMON</u>	
<u>84</u>	<u>86 03</u>		<u>LDA A #03</u>	<u>RESET:</u>	
<u>86</u>	<u>B7 F4 01</u>		<u>STA A CTRLA</u>	<u>ACIA (a)</u>	
<u>89</u>	<u>B7 F4 03</u>		<u>STA A CTRLB</u>	<u>ACIA (b)</u>	
<u>8C</u>	<u>6F F0 F0</u>	<u>SWI</u>	<u>STS PSTACK</u>	<u>SAVE PROGS. STACK PTR.</u>	
<u>8F</u>	<u>8E F0 00</u>	<u>START</u>	<u>LDS #STACK</u>	<u>SET UP MINIMONS STACK</u>	
<u>92</u>	<u>86 01</u>		<u>LDA A #01</u>	<u>SET UP</u>	
<u>94</u>	<u>B7 F4 01</u>		<u>STA A CTRLA</u>	<u>ACIA (a)</u>	
<u>97</u>	<u>BD FC 76</u>		<u>JSR STRING</u>	<u>PRINT:</u>	
<u>9A</u>	<u>00 0A 00</u>			<u>CR, LF, NULL, *</u>	
<u>9D</u>	<u>2A FF</u>				
<u>9F</u>	<u>BD FC 00</u>		<u>JSR RD</u>	<u>GET COMMAND</u>	
<u>A2</u>	<u>81 53</u>		<u>CMP A #53</u>		
<u>A4</u>	<u>27 93</u>		<u>BEQ BR.PTSET</u>	<u>IS IT S</u>	
<u>A6</u>	<u>81 48</u>		<u>CMP A #48</u>		
<u>A8</u>	<u>27 89</u>		<u>BEQ HEADER</u>	<u>" H</u>	
<u>AA</u>	<u>81 50</u>		<u>CMP A #50</u>		
<u>AC</u>	<u>26 03</u>		<u>BNE 1</u>	<u>" P</u>	
<u>AE</u>	<u>7E FC 08</u>		<u>JMP PUNCH</u>		
<u>B1</u>	<u>81 4C</u>	<u>1</u>	<u>CMP A #4C</u>		
<u>B3</u>	<u>26 03</u>		<u>BNE 2</u>	<u>" L</u>	
<u>B5</u>	<u>7E F0 07</u>		<u>JMP LOAD</u>		
<u>B8</u>	<u>81 52</u>	<u>2</u>	<u>CMP A #52</u>		
<u>BA</u>	<u>26 03</u>		<u>BNE 3</u>	<u>" R</u>	
<u>BC</u>	<u>7E F0 36</u>		<u>JMP REGPR</u>		
<u>BF</u>	<u>81 42</u>	<u>3</u>	<u>CMP A #42</u>		
<u>C1</u>	<u>26 03</u>		<u>BNE 4</u>	<u>" B</u>	
<u>C3</u>	<u>7E F0 62</u>		<u>JMP BLOCKM</u>		
<u>C6</u>	<u>81 5A</u>	<u>4</u>	<u>CMP A #5A</u>		
<u>C8</u>	<u>26 03</u>		<u>BNE 5</u>	<u>" Z</u>	
<u>CA</u>	<u>7E F0 E2</u>		<u>JMP Z</u>		
<u>CD</u>	<u>81 58</u>	<u>5</u>	<u>CMP A #58</u>		
<u>CF</u>	<u>26 03</u>		<u>BNE 6</u>	<u>" X</u>	
<u>D1</u>	<u>7E FE 1C</u>		<u>JMP XX</u>		

